

Predicting NBA Game Outcomes Using Betting Data

1. Introduction

This project analyzes historical game and betting data from the NBA to investigate whether game outcomes can be predicted using team statistics and sportsbook betting lines. The [dataset](#) spans from October 2007 to June 2025 and is sourced from the ESPN API via Kaggle.

The primary objective is to develop a predictive model that estimates the probability of a team winning a game. These probabilities are then compared against sportsbook implied odds to determine whether any consistent betting advantage exists.

2. Dataset Overview

The dataset contains detailed information on NBA games, including:

- Game outcomes (scores, winners)
- Betting lines (moneyline odds, spreads, favoured team)
- Team performance metrics
- Feature-engineered variables (rolling averages, differences between teams)

The dataset is updated regularly and distributed under a **CC0 (public domain)** license, allowing unrestricted use.

The data used in this project is pulled from the file `nba_2008-2025.csv`.

3. Research Questions

This project aims to answer the following:

1. Do home teams win more often than away teams, and is this reflected in betting odds?
 2. How often do sportsbooks correctly predict game outcomes?
 3. Which team statistics best predict game results?
 4. **Is a competitive advantage over the sportsbooks possible?**
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4. Methodology

4.1 Problem Framing

Rather than treating prediction as a classification problem (win/loss), this project uses a **regression-based approach**, where the model outputs:

- A probability that a team will win (between 0 and 1)

This allows direct comparison with sportsbook **implied probabilities**, which are derived from betting odds.

4.2 Data Pipeline

Data Acquisition

- Dataset downloaded as CSV from Kaggle
- Original data sourced from the ESPN API

ETL / Data Cleaning

- Cleaning performed in Excel:
 - Removed irrelevant columns from the 27-column dataset
 - Rows with missing values were not outright removed, as there is still useful data in those rows
 - Constructed a data dictionary documenting all transformations
- Cleaned dataset exported as CSV
- Data extension performed with Python:
 - Derived columns based on rolling averages
 - Documented changes in the data dictionary
- Extended dataset exported as CSV

Data Storage

- Data stored in well-structured CSV files. These files are directly utilized in the R and Python code in the project
 - Cleaned data uploaded into [Google BigQuery](#)
 - Enables scalable querying and structured data access
 - BigQuery was not meaningfully used, but CSV usage was more intuitive for the scope of this project.
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4.3 Exploratory Data Analysis (EDA)

EDA was conducted in R.

Data for missingness:

column	n_missing	pct_missing	n_complete
moneyline_away	3028	15.58	16411
moneyline_home	3028	15.58	16411
points_against_home	252	1.30	19187
points_for_home	252	1.30	19187
points_against_away	228	1.17	19211
points_for_away	228	1.17	19211
points_against_last10_home	17	0.09	19422
points_for_last10_home	17	0.09	19422
points_against_last10_away	13	0.07	19426
points_for_last10_away	13	0.07	19426
h2_spread	2	0.01	19437
id_spread	2	0.01	19437
spread	2	0.01	19437
away	0	0.00	19439
away_record	0	0.00	19439
date	0	0.00	19439
days_rest_away	0	0.00	19439
days_rest_home	0	0.00	19439
h2_total	0	0.00	19439
home	0	0.00	19439
home_record	0	0.00	19439
id_total	0	0.00	19439
implied_odds_away	0	0.00	19439
implied_odds_home	0	0.00	19439
last_game_ot_away	0	0.00	19439
last_game_ot_home	0	0.00	19439
last10_win_loss_away	0	0.00	19439
last10_win_loss_home	0	0.00	19439
ot_away	0	0.00	19439
ot_home	0	0.00	19439
profit_moneyline_away	0	0.00	19439
profit_moneyline_home	0	0.00	19439
q1_away	0	0.00	19439
q1_home	0	0.00	19439
q2_away	0	0.00	19439
q2_home	0	0.00	19439
q3_away	0	0.00	19439
q3_home	0	0.00	19439
q4_away	0	0.00	19439
q4_home	0	0.00	19439
regular	0	0.00	19439
score_away	0	0.00	19439
score_home	0	0.00	19439
season	0	0.00	19439
total	0	0.00	19439
west_conference_away	0	0.00	19439
west_conference_home	0	0.00	19439
whos_favored	0	0.00	19439
win_loss_away	0	0.00	19439
win_loss_home	0	0.00	19439

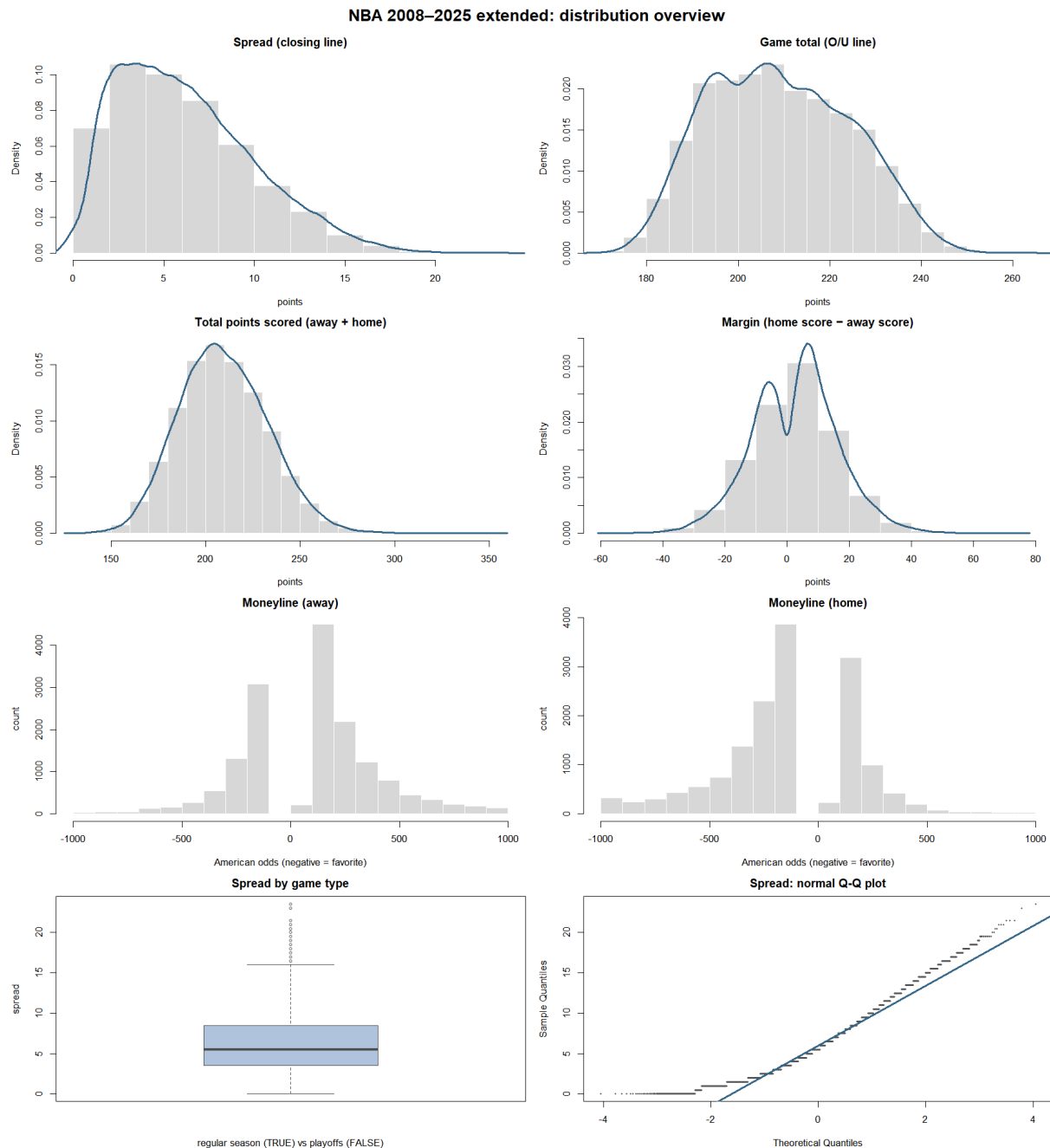
Summary stats:

column	n_valid	mean	sd	min	median	max	n_missing
days_rest_away	19439	1.0106	0.9587	0.00	1.0000	5.0000	0
days_rest_home	19439	1.2225	0.9600	0.00	1.0000	5.0000	0
h2_spread	19437	0.5192	0.5268	0.00	1.0000	2.0000	2
h2_total	19439	0.5155	0.5219	0.00	1.0000	2.0000	0
id_spread	19437	0.5192	0.5268	0.00	1.0000	2.0000	2
id_total	19439	0.5155	0.5219	0.00	1.0000	2.0000	0
implied_odds_away	19439	0.3590	0.2470	0.00	0.3571	0.9900	0
implied_odds_home	19439	0.5170	0.2940	0.00	0.5745	0.9924	0
last_game_ot_away	19439	0.0565	0.2310	0.00	0.0000	1.0000	0
last_game_ot_home	19439	0.0544	0.2269	0.00	0.0000	1.0000	0
moneyline_away	16411	130.7922	446.6094	-9900.00	150.0000	6500.0000	3028
moneyline_home	16411	-267.2225	708.3878	-13000.00	-170.0000	3000.0000	3028
ot_away	19439	0.6695	3.0442	0.00	0.0000	44.0000	0
ot_home	19439	0.7079	3.1777	0.00	0.0000	41.0000	0
points_against_away	19211	104.0054	7.7428	71.00	103.2700	136.0000	228
points_against_home	19187	104.0464	7.7651	70.00	103.2800	143.0000	252
points_against_last10_away	19426	104.6921	8.2759	74.00	104.2000	132.1000	13
points_against_last10_home	19422	104.6937	8.2711	80.30	104.1000	130.5000	17
points_for_away	19211	104.0672	7.8279	71.00	103.3200	143.0000	228
points_for_home	19187	103.9947	7.8018	59.00	103.2500	140.0000	252
points_for_last10_away	19426	104.7417	8.3290	79.90	104.0000	132.4000	13
points_for_last10_home	19422	104.6453	8.2741	74.00	104.1000	133.4000	17
profit_moneyline_away	19439	-13.1681	69.8495	-487.06	0.0000	323.0600	0
profit_moneyline_home	19439	-12.8788	70.0322	-467.06	0.0000	333.0600	0
q1_away	19439	25.9049	5.9174	7.00	26.0000	55.0000	0
q1_home	19439	26.7770	5.9347	5.00	27.0000	55.0000	0
q2_away	19439	25.9647	5.8870	4.00	26.0000	49.0000	0
q2_home	19439	26.6227	5.8732	6.00	26.0000	51.0000	0
q3_away	19439	25.5929	5.9767	5.00	25.0000	52.0000	0
q3_home	19439	26.3051	6.0510	6.00	26.0000	55.0000	0
q4_away	19439	25.3018	5.9777	5.00	25.0000	48.0000	0
q4_home	19439	25.6597	5.8591	7.00	26.0000	50.0000	0
regular	19439	1.0000	0.0000	1.00	1.0000	1.0000	0
score_away	19439	103.4338	13.6160	54.00	103.0000	176.0000	0
score_home	19439	106.0724	13.5200	59.00	106.0000	175.0000	0
season	19439	2016.0495	5.3055	2008.00	2016.0000	2025.0000	0
spread	19437	6.2215	3.7356	0.00	5.5000	23.5000	2
total	19439	209.1229	15.3520	170.50	208.0000	262.5000	0
west_conference_away	19439	0.5000	0.5000	0.00	1.0000	1.0000	0
west_conference_home	19439	0.5000	0.5000	0.00	1.0000	1.0000	0

There is, of course, a significant number of moneyline statistics that are missing. As mentioned above, though, it is important to keep these rows, as they are still important to use as predictors for future games. What we can learn from this is that we need to remember to properly handle missing values when we are building our predictive models.

The summary statistics are not too useful for this specific task. Since we are considering head-to-head matchups, these absolute values will not tell us much. Predictions will be comparative.

Distribution graphs:



- The home team is favoured to win more often than the away team.
- The home team wins the game more often than the away team.

- A majority of games are predicted to have a margin of victory between 4 and 9 points, meaning that blowout victories are not commonly predicted.
 - As previously mentioned, since predictions for the winners of individual games are comparative rather than absolute, distribution graphs will not be significant.
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4.4 Feature Selection and Model Development

Logistic regression was the baseline for all analyses, with the goal being to calculate the probability between 0 and 1 that a team will win a game. The following feature selection techniques were used to develop their own models:

- BIC Stepwise Regression
 - Filter Univariate Selection
 - L1 Regularization (lenient lambda)
 - L1 Regularization (lambda.min)
 - PCA Loading
 - Ridge Regression
 - Stability LASSO
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4.5 Training and Testing

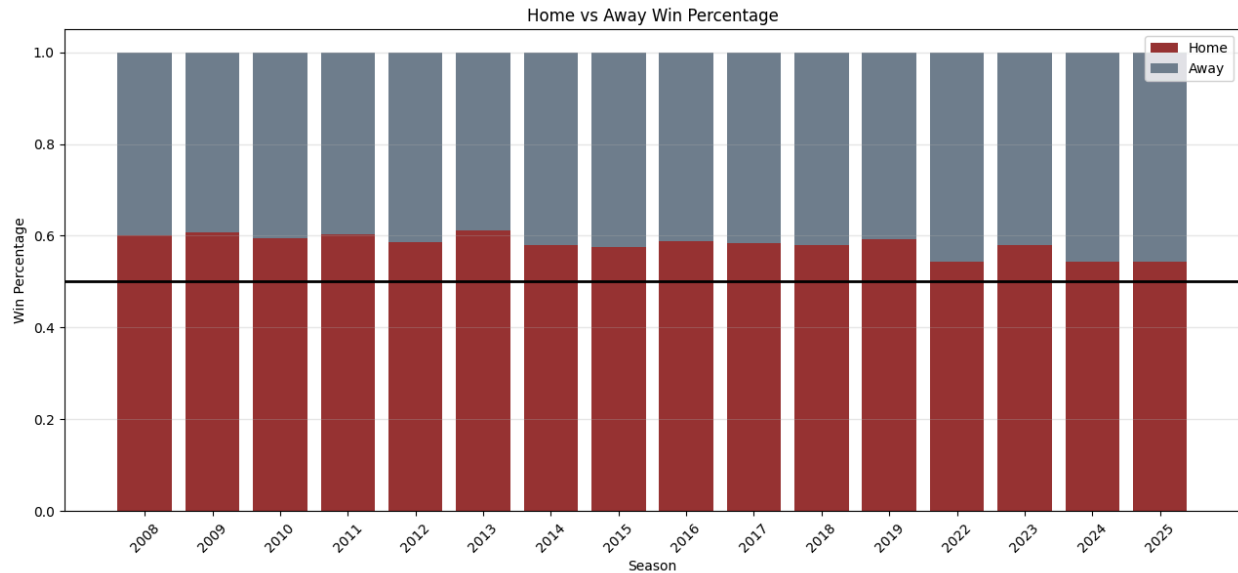
- A rolling approach is used for training vs. testing data when the models are being built. Example: Data from 2005 is used to predict 2006, data from 2005 and 2006 is used to predict 2007, data from 2005, 2006, and 2007 is used to predict 2008, etc.
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5. Results

5.1 Home vs Away Performance

Question: Do home teams win more often than away teams, and is this reflected in betting odds?

- Home teams win approximately **55-60%** of games
- This advantage has **declined over time**



This suggests that while home-court advantage exists, it is weakening.

5.2 Sportsbook Prediction Accuracy

Question: How often do sportsbooks correctly predict game outcomes?

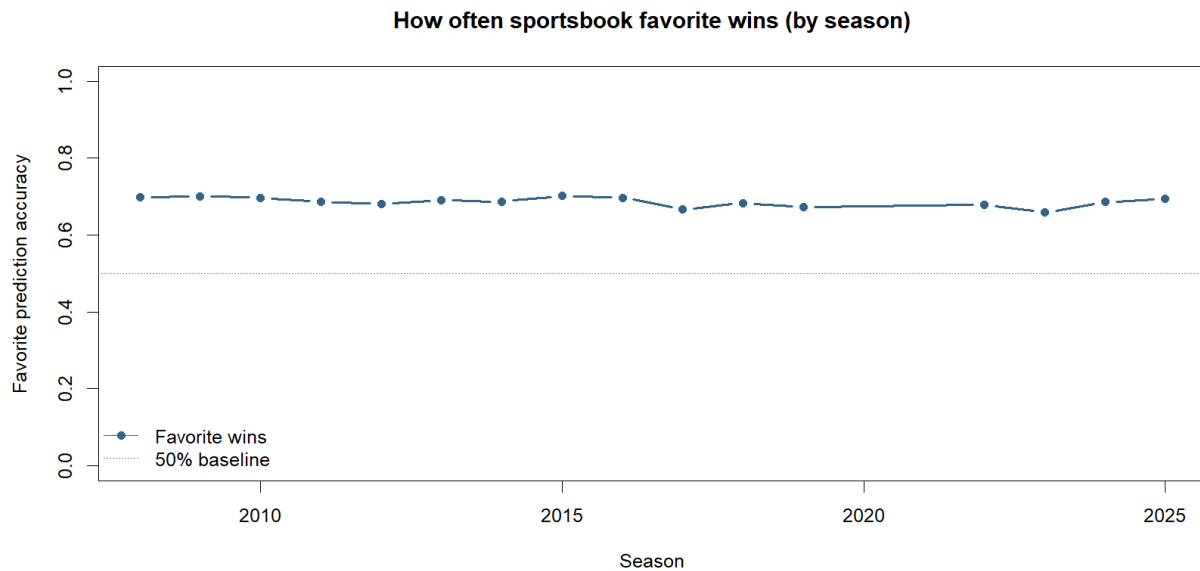
- Sportsbooks correctly predict winners in the majority of games
- The favoured team (lower odds) wins most often: 68.65% of the time

However:

- Upsets occur regularly: 31.35% of the time
- Odds are not perfectly calibrated

Note:

- A gap can be seen in the 2020 and 2021 seasons. As mentioned in the data dictionary found in GitHub, those seasons are not considered during this project due to the COVID-19 pandemic. Because of COVID restrictions, home court advantage essentially disappeared, and the game of basketball was different as a whole.



5.3 Key Predictors

Question: Which team statistics best predict game results?

This analysis found that home court advantage is the most consistent predictor of game outcomes in the NBA, as home teams win a majority of games across all seasons analyzed.

However, team-level statistics available in this dataset do not provide sufficient detail to reliably predict outcomes on their own. While point totals describe what happened in a game, they do not capture the underlying factors that drive results.

Additionally, betting market data suggests that many predictive factors are already incorporated into point spreads and odds. The fact that favourites win more often, but not always, indicates that sportsbooks effectively model team strength and game conditions, limiting the ability to gain predictive advantage using basic statistics alone.

Overall, while simple metrics like scoring and win rates offer descriptive insights, more advanced statistics are required to build strong predictive models of NBA game outcomes.

5.4 Competitive Advantage

Question: Is a competitive advantage over the sportsbooks possible?

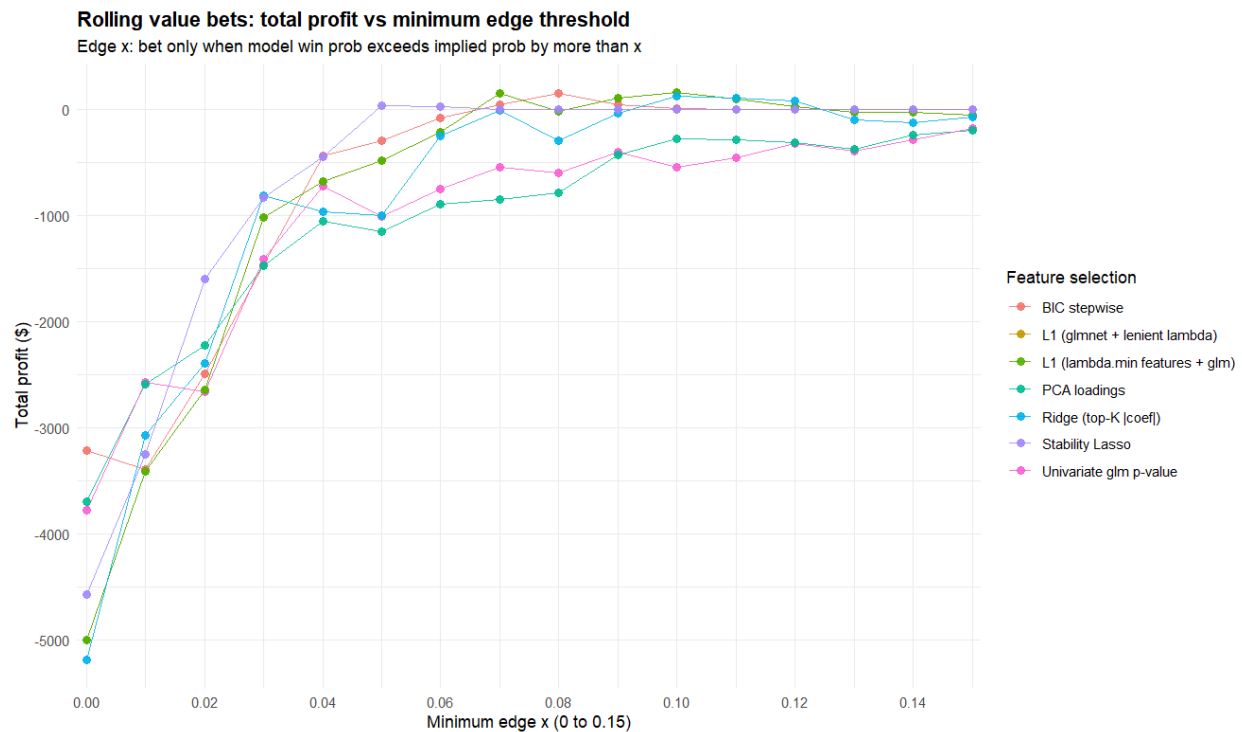
predictor	n	n_distinct	aic	pseudo_r2	lrt_p_value	coef_first_term	note
moneyline_away	16410	619	19396.83	0.12756	0.000000	0.002807	NA
moneyline_home	16410	630	19500.94	0.12287	0.000000	-0.002412	NA
whos_favored	19438	2	24135.12	0.08660	0.000000	1.501188	NA
implied_odds_away	19438	619	24905.69	0.05743	0.000000	-2.382036	NA
implied_odds_home	19438	624	24964.37	0.05521	0.000000	1.940732	NA
last10_win_loss_home	19438	57	25495.76	0.03926	0.000000	NA	NA
last10_win_loss_away	19438	57	25708.75	0.03120	0.000000	NA	NA
spread	19436	46	25720.37	0.02646	0.000000	0.107383	NA
h2_spread	19436	3	25945.58	0.01794	0.000000	0.609338	NA
id_spread	19436	3	25945.58	0.01794	0.000000	0.609338	NA
home	19438	30	26017.49	0.01747	0.000000	NA	NA
away	19438	30	26183.31	0.01119	0.000000	NA	NA
points_against_last10_home	19421	535	26151.51	0.00943	0.000000	-0.027919	NA
points_against_home	19186	3351	25842.37	0.00934	0.000000	-0.029570	NA
points_for_away	19210	3309	25921.08	0.00756	0.000000	-0.026358	NA
points_for_last10_away	19425	535	26219.75	0.00704	0.000000	-0.023891	NA
profit_moneyline_home	19438	11335	26240.28	0.00691	0.000000	0.002629	NA
profit_moneyline_away	19438	11325	26270.72	0.00576	0.000000	-0.002619	NA
points_for_last10_home	19421	525	26301.89	0.00374	0.000000	0.017552	NA
points_for_home	19186	3296	25995.81	0.00345	0.000000	0.017892	NA
points_against_last10_away	19425	533	26331.11	0.00282	0.000000	0.015224	NA
points_against_away	19210	3310	26060.48	0.00222	0.000000	0.014440	NA
west_conference_home	19438	2	26376.76	0.00175	0.000000	0.197715	NA
date_num	19438	2572	26398.00	0.00094	0.000001	-0.000038	NA
season	19438	16	26398.13	0.00094	0.000001	-0.013643	NA
days_rest_away	19438	6	26409.88	0.00049	0.000306	-0.054551	NA
west_conference_away	19438	2	26410.91	0.00045	0.000533	-0.100767	NA
last_game_ot_home	19438	2	26416.39	0.00025	0.010656	-0.162638	NA
days_rest_home	19438	6	26416.90	0.00023	0.014207	0.037305	NA
total	19438	163	26417.59	0.00020	0.021066	-0.002185	NA
last_game_ot_away	19438	2	26420.39	0.00010	0.112320	0.100525	NA
h2_total	19438	3	26421.50	0.00005	0.234371	0.033143	NA
id_total	19438	3	26421.50	0.00005	0.234371	0.033143	NA
regular	19438	1	26420.91	0.00000	NA	NA	NA
win_loss_away	19438	2471	NA	NA	NA	NA	skipped: 2471 distinct strings (> 100)

predictor	total_profit	n_bets
away_record	0.00	0
home_record	0.00	0
win_loss_away	0.00	0
win_loss_home	0.00	0
moneyline_home	-4204.54	12137
moneyline_away	-4576.80	11072
season	-4847.81	13180
points_against_last10_home	-5319.45	14187
home	-5854.79	14163
points_for_home	-5905.55	14026
spread	-5947.08	14733
last10_win_loss_home	-6069.10	14097
west_conference_home	-6121.67	14304
date_num	-6136.22	14311
points_against_last10_away	-6315.86	14148
regular	-6331.53	14307
points_for_last10_home	-6341.66	14227
last_game_ot_home	-6352.20	14286
days_rest_away	-6408.20	14276
days_rest_home	-6412.82	14319
last_game_ot_away	-6417.29	14301
points_against_away	-6482.25	14030
h2_total	-6529.65	14299
id_total	-6529.65	14299
points_for_last10_away	-6696.68	14232
points_against_home	-6736.06	14003
total	-6761.88	14295
away	-6899.65	14172
west_conference_away	-6928.65	14310
points_for_away	-6978.87	14073
whos_favored	-7280.95	13542
last10_win_loss_away	-8405.40	14126

The baseline analysis involved using single predictors and analyzing stats related to their linear regressions. The left graphic ranks each predictor based on how useful it is by itself, by R^2 . This shows us that the most accurate single predictors are taken from the sportsbooks themselves in the form of the moneyline and the implied odds that are generated from that moneyline. The next most useful predictor appears to be the team's record over the last 10 games.

The right graphic shows how much money would be made if \$10 was placed every time there was an implied advantage using that metric. For example, if the implied odds are 0.63 and the model gives a 0.65 chance of a victory, it puts \$10 on that game. If the model gives anything less than 0.63, then it would not try to put \$10 on that game.

Here is a comparison between all of the predictor models outlined in Section 4.4:



To determine if a mathematical edge exists, we evaluated seven distinct feature selection techniques. Unlike a standard model that predicts every game, this simulation applies a "Minimum Edge Threshold" (the x-axis). This means the model only "placed a bet" when its calculated win probability exceeded the sportsbook's implied probability by a specific margin (e.g., an edge of 0.05 or 5%).

Key Observations:

- As the threshold moves to the right, the number of bets placed decreases. While one might expect higher certainty to yield higher ROI, the graph shows that even with a high edge requirement (0.10+), most models still hover around the "zero-line."
- The primary reason these lines struggle to stay above zero is the sportsbook "vig." To be profitable, a model doesn't just need to be "right" more than 50% of the time; it must be right often enough to overcome the transaction cost built into the odds.

The convergence of all seven techniques toward the zero-line, regardless of the mathematical approach used, serves as strong evidence of market efficiency. While feature selection helps optimize the model, no single statistical "twist" was sufficient to generate a reliable, scalable profit against NBA betting lines.

6. Discussion

6.1 Limitations

There are some limitations with the approach that we took.

- **Scale:** We are 3 students doing a class project going up against several data analysts with supercomputers on their side. This was an uphill battle from the start
 - **Data Availability:** We are using an open-source data set. The oddsmakers have access to deeper data, including which players are available on a given night, as well as some data that we do not even know exists.
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6.2 Areas for Optimism

Sportsbooks are generally efficient because they adjust odds based on market behaviour. This means that if the general public is inaccurately predicting that a team will win a game, there could still be profitability with betting on the other team to win.

6.3 Next Steps

If there is any hope of attempting to gain a competitive advantage over sportsbooks, one or more of the following will be necessary:

- Insights into how exactly odds are calculated by the oddsmakers
- More topical data, such as which players are active for a specific game
- Access to market data, such as how many bets have been made on either team

Another consideration for the next step is to attempt a similar analysis on a different sport. Different sports inherently have different levels of predictability. For example, the favoured team in basketball will consistently play with a significant advantage, while a more entropic sport like hockey could have less predictable outcomes. There could be a potential path to profitability by extending the analysis to other sports.

7. Conclusion

This project successfully developed a scalable data pipeline to test the efficiency of NBA betting markets. While our models achieved high descriptive accuracy, they failed to generate a statistically significant edge over sportsbook implied odds. The consistent "zero-line"

performance across all modelling techniques suggests that the NBA betting market is highly efficient, likely incorporating the very team-level statistics and home-court advantages identified in our EDA. Ultimately, while team metrics can describe game outcomes with high reliability, the "vig" (sportsbook margin) and the inclusion of real-time variables by oddsmakers create a barrier to entry that basic historical data cannot overcome.

8. Ethical Considerations

Although the dataset is publicly available (CC0 license), there are ethical concerns related to sports betting.

This project:

- Does **not promote gambling**
 - Is conducted strictly for **educational and analytical purposes**
 - Acknowledges the financial risks associated with betting
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9. AI Disclaimer

- **Human Oversight:** All architectural decisions, feature engineering logic and the interpretation of model results are the original work of the authors.
 - **Scope of Use:** AI was used for Python and R code generation.
 - **Tooling:** The Cursor IDE was used to generate code during this project.
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10. Published Location

The published project can be found at <https://github.com/maximstorozhuk/DataPipeline>